



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 03

Title: Implement control structures and form validation; create a grading system based on user-entered marks.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<html>

<head>

<title>JS Practical - Multi-Subject Grading System</title>

<style>

body {

    font-family: Arial, sans-serif;

    max-width: 480px;

    margin: 30px auto;

    padding: 20px;

    border: 1px solid #ccc;

    border-radius: 8px;

    box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);

}

.form-group {

    margin-bottom: 15px;

}
```



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```
label {  
  
    display: block;  
  
    margin-bottom: 4px;  
  
    font-weight: bold;  
  
}  
  
input {  
  
    width: 100%;  
  
    padding: 8px;  
  
    box-sizing: border-box;  
  
    border: 1px solid #ccc;  
  
    border-radius: 4px;  
  
    outline: none;  
  
}  
  
/* Highlight invalid input fields */  
  
input.invalid-input {  
  
    border: 2px solid #dc3545;  
  
    background-color: #fff8f8;  
  
}  
  
/* Style inline error text under inputs */  
  
.field-error {  
  
    color: #dc3545;  
  
    font-size: 13px;
```



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```
margin-top: 4px;

display: block;

min-height: 18px; /* Reserves space so layout doesn't jump */
```

```
}
```

```
button {
```

```
width: 100%;

padding: 10px;

background-color: #007bff;

color: white;

border: none;

border-radius: 4px;

cursor: pointer;

font-size: 16px;

margin-top: 10px;

}
```

```
button:hover {
```

```
background-color: #0056b3;

}
```

```
#result {
```

```
margin-top: 20px;

padding: 12px;

border-radius: 4px;
```



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```
display: none;

background-color: #d4edda;

color: #155724;

border: 1px solid #c3e6cb;

}

</style>

</head>

<body>

<h2>Grading System</h2>

<form id="gradingForm" onsubmit="calculateAverageGrade(event)" novalidate>

  <div class="form-group">

    <label for="studentName">Student Name:</label>

    <input type="text" id="studentName" placeholder="e.g. John Doe">

    <span class="field-error" id="nameError"></span>

  </div>

  <div class="form-group">

    <label for="math">Math Marks (0 - 100):</label>

    <input type="number" id="math" placeholder="e.g. 85">
```



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</div>

<div class="form-group">

<label for="science">Science Marks (0 - 100):</label>

<input type="number" id="science" placeholder="e.g. 90">

</div>

<div class="form-group">

<label for="english">English Marks (0 - 100):</label>

<input type="number" id="english" placeholder="e.g. 78">

</div>

<button type="submit">Calculate Overall Grade</button>

</form>

<div id="result"></div>

<script>

function calculateAverageGrade(event) {



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```
event.preventDefault();
```

```
// Track whether the form is completely valid
```

```
let isValid = true;
```

```
// Reset previous result box
```

```
const resultDiv = document.getElementById("result");
```

```
resultDiv.style.display = "none";
```

```
// -----
```

```
// Helper Function: Validate an individual field
```

```
// -----
```

```
function validateField(inputId, errorId, fieldName, isSubject = true) {
```

```
    const inputElement = document.getElementById(inputId);
```

```
    const errorElement = document.getElementById(errorId);
```

```
    const value = inputElement.value.trim();
```

```
// Clear previous styles/messages
```

```
inputElement.classList.remove("invalid-input");
```

```
errorElement.textContent = "";
```

```
// Rule 1: Empty text validation
```



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```
if (value === "") {  
    inputElement.classList.add("invalid-input");  
    errorElement.textContent = `Please enter ${fieldName}.`;   
    isValid = false;  
    return;  
}  
  
// Rule 2: Numeric & Range validation (for subject marks)  
if (isSubject) {  
    const score = parseFloat(value);  
    if (isNaN(score) || score < 0 || score > 100) {  
        inputElement.classList.add("invalid-input");  
        errorElement.textContent = `Enter valid ${fieldName} (0–100).`;   
        isValid = false;  
    }  
}  
  
// -----  
// 1. RUN VALIDATIONS FOR ALL FIELDS INDEPENDENTLY  
// -----  
  
validateField("studentName", "nameError", "student's name", false);
```



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```
validateField("math", "mathError", "Math marks", true);

validateField("science", "scienceError", "Science marks", true);

validateField("english", "englishError", "English marks", true);


// Stop calculation if any field failed validation
if (!isValid) return;


// -----

// 2. CALCULATE GRADE (Runs only if all fields are valid)
// -----

const math = parseFloat(document.getElementById("math").value);
const science = parseFloat(document.getElementById("science").value);
const english = parseFloat(document.getElementById("english").value);
const name = document.getElementById("studentName").value.trim();


const totalMarks = math + science + english;
const average = (totalMarks / 3).toFixed(2);


let grade = "";
let status = "";


if (average >= 90) {
```




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```
grade = "A+"; status = "Passed with Distinction";

} else if (average >= 80) {

    grade = "A"; status = "Passed";

} else if (average >= 70) {

    grade = "B"; status = "Passed";

} else if (average >= 60) {

    grade = "C"; status = "Passed";

} else if (average >= 50) {

    grade = "D"; status = "Passed";

} else {

    grade = "F"; status = "Failed";

}

// Display result summary

resultDiv.style.display = "block";

resultDiv.innerHTML = `

    <strong>Result Summary:</strong><br>

    Name: ${name}<br>

    Math: ${math} | Science: ${science} | English: ${english}<br>

    Total: ${totalMarks} / 300<br>

    Average: <strong>${average}%</strong><br>

    Overall Grade: <strong>${grade}</strong> (${status})
```



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```
`;  
}  
  
</script>
```

```
</body>
```

```
</html>
```

Output:

Grading System

Student Name:
Aishwarya Brahmanekar

Math Marks (0 - 100):
95

Science Marks (0 - 100):
89

English Marks (0 - 100):
97

Calculate Overall Grade

Result Summary:
Name: Aishwarya Brahmanekar
Math: 95 | Science: 89 | English: 97
Total: 281 / 300
Average: 93.67%
Overall Grade: A+ (Passed with Distinction)



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Case Study No.: 03

Title: Implement control structures and form validation; create a grading system based on user-entered marks.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<html>
```

```
<head>
```

```
<title>JS Practical - Multi-Subject Grading System</title>
```

```
<style>
```

```
body {
```

```
    font-family: Arial, sans-serif;
```

```
    max-width: 480px;
```

```
    margin: 30px auto;
```

```
    padding: 20px;
```

```
    border: 1px solid #ccc;
```

```
    border-radius: 8px;
```

```
    box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
```

```
}
```

```
.form-group {
```



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margin-bottom: 15px;

}

label {

display: block;

margin-bottom: 4px;

font-weight: bold;

}

input {

width: 100%;

padding: 8px;

box-sizing: border-box;

border: 1px solid #ccc;

border-radius: 4px;

outline: none;

}

/* Highlight invalid input fields */

input.invalid-input {

border: 2px solid #dc3545;

background-color: #fff8f8;



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}

/* Style inline error text under inputs */

.field-error {

color: #dc3545;

font-size: 13px;

margin-top: 4px;

display: block;

min-height: 18px; /* Reserves space so layout doesn't jump */

}

button {

width: 100%;

padding: 10px;

background-color: #007bff;

color: white;

border: none;

border-radius: 4px;

cursor: pointer;

font-size: 16px;

margin-top: 10px;



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}

button:hover {

background-color: #0056b3;

}

#result {

margin-top: 20px;

padding: 12px;

border-radius: 4px;

display: none;

background-color: #d4edda;

color: #155724;

border: 1px solid #c3e6cb;

}

</style>

</head>

<body>

<h2>Grading System</h2>



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```
<form id="gradingForm" onsubmit="calculateAverageGrade(event)" novalidate>
```

```
<div class="form-group">
```

```
<label for="studentName">Student Name:</label>
```

```
<input type="text" id="studentName" placeholder="e.g. John Doe">
```

```
<span class="field-error" id="nameError"></span>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="math">Math Marks (0 - 100):</label>
```

```
<input type="number" id="math" placeholder="e.g. 85">
```

```
<span class="field-error" id="mathError"></span>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="science">Science Marks (0 - 100):</label>
```

```
<input type="number" id="science" placeholder="e.g. 90">
```

```
<span class="field-error" id="scienceError"></span>
```

```
</div>
```



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```
<div class="form-group">
```

```
<label for="english">English Marks (0 - 100):</label>
```

```
<input type="number" id="english" placeholder="e.g. 78">
```

```
<span class="field-error" id="englishError"></span>
```

```
</div>
```

```
<button type="submit">Calculate Overall Grade</button>
```

```
</form>
```

```
<div id="result"></div>
```

```
<script>
```

```
function calculateAverageGrade(event) {
```

```
    event.preventDefault();
```

```
    // Track whether the form is completely valid
```

```
    let isValid = true;
```




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```
// Reset previous result box

const resultDiv = document.getElementById("result");

resultDiv.style.display = "none";


// -----

// Helper Function: Validate an individual field

// -----

function validateField(inputId, errorId, fieldName, isSubject = true) {

    const inputElement = document.getElementById(inputId);

    const errorElement = document.getElementById(errorId);

    const value = inputElement.value.trim();

    // Clear previous styles/messages

    inputElement.classList.remove("invalid-input");

    errorElement.textContent = "";

    // Rule 1: Empty text validation

    if (value === "") {

        inputElement.classList.add("invalid-input");
```



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```
errorElement.textContent = `Please enter ${fieldName}.`;
```

```
isValid = false;
```

```
return;
```

```
}
```

```
// Rule 2: Numeric & Range validation (for subject marks)
```

```
if (isSubject) {
```

```
    const score = parseFloat(value);
```

```
    if (isNaN(score) || score < 0 || score > 100) {
```

```
        inputElement.classList.add("invalid-input");
```

```
        errorElement.textContent = `Enter valid ${fieldName} (0–100).`;
```

```
        isValid = false;
```

```
    }
```

```
}
```

```
}
```

```
// -----
```

```
// 1. RUN VALIDATIONS FOR ALL FIELDS INDEPENDENTLY
```

```
// -----
```



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```
validateField("studentName", "nameError", "student's name", false);
```

```
validateField("math", "mathError", "Math marks", true);
```

```
validateField("science", "scienceError", "Science marks", true);
```

```
validateField("english", "englishError", "English marks", true);
```

```
// Stop calculation if any field failed validation
```

```
if (!isValid) return;
```

```
// -----
```

```
// 2. CALCULATE GRADE (Runs only if all fields are valid)
```

```
// -----
```

```
const math = parseFloat(document.getElementById("math").value);
```

```
const science = parseFloat(document.getElementById("science").value);
```

```
const english = parseFloat(document.getElementById("english").value);
```

```
const name = document.getElementById("studentName").value.trim();
```

```
const totalMarks = math + science + english;
```

```
const average = (totalMarks / 3).toFixed(2);
```



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```
let grade = "";

let status = "";

let scholarship = "";

if (average >= 95) {

    grade = "A+"; status = "Passed with Distinction"; scholarship = "Eligible for Platinum
Scholarship";

} else if (average >= 85) {

    grade = "A"; status = "Passed"; scholarship = "Eligible for Gold Scholarship";

} else if (average >= 70) {

    grade = "B"; status = "Passed"; scholarship = "Eligible for Silver Scholarship";

} else if (average >= 50) {

    grade = "C"; status = "Passed"; scholarship = "Eligible for Bronze Scholarship";

} else if (average < 50) {

    grade = "D"; status = "Passed"; scholarship = "No Scholarship";

} else {

    grade = "F"; status = "Failed";

}
```



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// Display result summary

```
resultDiv.style.display = "block";
```

```
resultDiv.innerHTML = `
```

```
    <strong>Result Summary:</strong><br>
```

```
    Name: ${name}<br>
```

```
    Math: ${math} | Science: ${science} | English: ${english}<br>
```

```
    Total: ${totalMarks} / 300<br>
```

```
    Average: <strong>${average}%</strong><br>
```

```
    Overall Grade: <strong>${grade}</strong> (${status}) ${scholarship}
```

```
`;
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

Output:



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127.0.0.1:5500/Aishwarya_Brahmankar_24070521212/Case%20Study%203/scholarship_grading_system.html

Grading System

Student Name:
Aishwarya Brahmankar

Math Marks (0 - 100):
95

Science Marks (0 - 100):
89

English Marks (0 - 100):
97

Calculate Overall Grade

Result Summary:
Name: Aishwarya Brahmankar
Math: 95 | Science: 89 | English: 97
Total: 281 / 300
Average: 93.67%
Overall Grade: A (Passed) Eligible for Gold Scholarship